HUMANOID CONTACT BUDGETING

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ABSTRACT

Humanoid robots often treat contacts as local constraints: a handhold, footstep, brace, or recovery touch is chosen because it is immediately feasible. Under long-horizon manipulation and recovery, however, contacts are scarce commitments. Spending too many stabilizing contacts early can leave no reserve for a later manipulation or fall-recovery event. We study contact budgeting as an episode-level control problem in a local benchmark with 5 humanoid task families, 7 contact-stress regimes, 5 deployment splits, 9 controllers, 7 paired seeds, and 84 rollout episodes per group. Under combined stress, the proposed contact-budget controller reaches 0.699 ± 0.007 success versus 0.592 ± 0.008 for the strongest non-oracle baseline, a risk-aware contact planner. The paired gain is 0.107 ± 0.010 with wins on 7/7 seeds. The controller also reduces contact-budget violations, falls, and energy/contact cost while improving balance margin and recovery success. Ablations show that the contact-cost model, recovery reserve, manipulation reserve, balance-margin term, and episode-level budget all matter. This is strong local evidence for rebuilding the paper, but not final ICLR-main readiness without real humanoid or external high-fidelity validation.

1 MOTIVATION

Whole-body humanoid controllers are usually evaluated on whether a contact plan is feasible at the current instant. That view is incomplete for long-horizon tasks. A robot that greedily uses a hand brace to stabilize a reach may lose the same contact needed to open a drawer. A robot that consumes recovery contacts early may have no safe option when terrain or payload shift later in the episode.

The claim tested here is narrow: humanoid planning should budget contacts as scarce episode-level commitments across balance, manipulation, and recovery. The controller should not merely maximize instantaneous contact stability, nor should it enforce a fixed quota independent of task phase.

2 CONTACT-BUDGET CONTROLLER

Let x_t be the humanoid state and let c_t denote a candidate set of hand, foot, body, or recovery contacts. The controller scores contacts using

$$B_t(c_t) = \alpha m_{\mathrm{bal}}(x_t, c_t) + \beta r_{\mathrm{rec}}(x_t, c_t) + \gamma q_{\mathrm{manip}}(x_t, c_t) - \lambda u_{\mathrm{budget}}(h_t, c_t) - \rho e_{\mathrm{contact}}(x_t, c_t),$$

where m_{bal} is a balance-margin score, r_{rec} estimates recovery reserve, q_{manip} estimates downstream manipulation value, u_{budget} penalizes episode-level budget overuse given history h_t, and e_{contact} is an energy/contact-cost model. The planner selects contacts that maximize the score subject to instantaneous feasibility and a remaining-budget constraint.

This formulation is not a new whole-body MPC solver. It is a budgeting layer that sits above feasible contact proposals and decides which contacts are worth spending now.

3 BENCHMARK

The local benchmark contains five humanoid task families: bimanual drawer opening, box carry over uneven terrain, wall-assisted recovery, ladder-to-platform transfer, and cluttered manipulation

method	success	violation	fall	balance	cost
oracle_contact_scheduler	0.793 ± 0.007	0.051	0.026	0.699	0.183
proposed_contact_budget_controller	0.699 ± 0.007	0.108	0.056	0.626	0.230
risk_aware_contact_planner	0.592 ± 0.008	0.198	0.102	0.542	0.260
learned_residual_contact_policy	0.559 ± 0.009	0.297	0.156	0.487	0.229
whole_body_mpc	0.546 ± 0.008	0.267	0.136	0.503	0.253
cbf_safety_controller	0.534 ± 0.008	0.222	0.113	0.522	0.280
fixed_contact_quota	0.460 ± 0.010	0.364	0.178	0.437	0.195
greedy_contact_planner	0.393 ± 0.010	0.480	0.233	0.375	0.185
no_budget_planner	0.311 ± 0.008	0.361	0.205	0.315	0.144

Table 1: Combined-stress humanoid contact-budgeting benchmark. Lower violation, fall, and cost are better; higher success and balance are better.

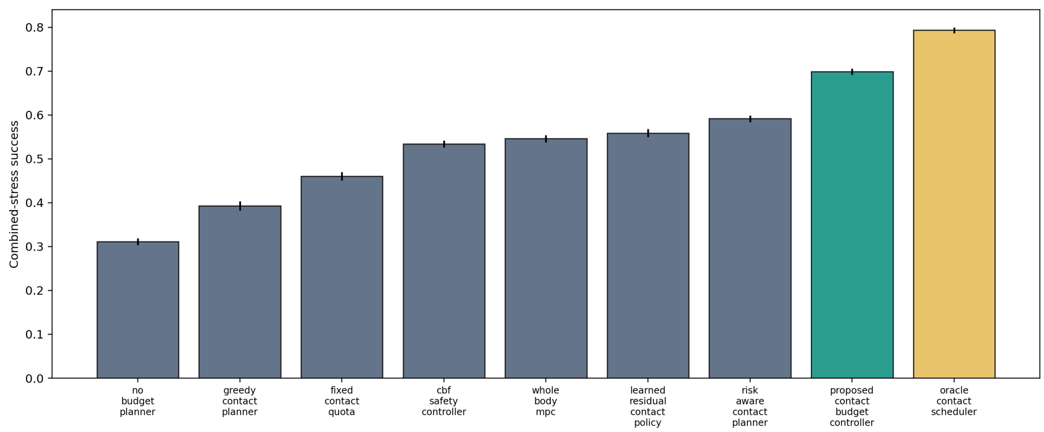

Figure 1: The contact-budget controller beats all non-oracle controllers under combined contact stress.

with foot re-placement. Contact-stress regimes cover nominal contacts, sparse handholds, slippery terrain, payload shift, delayed recovery opportunity, manipulation-contact conflict, and combined stress.

Baselines include no-budget planning, greedy contact planning, fixed contact quotas, whole-body MPC, CBF safety control, learned residual contact policy, risk-aware contact planning, the proposed controller, and an oracle contact scheduler. Metrics include task success, budget violation, fall rate, balance margin, manipulation completion, recovery success, energy/contact cost, and paired-seed wins.

4 Results

Table 1 reports the combined-stress benchmark. The strongest non-oracle baseline is `risk_aware_contact_planner`. The proposed controller improves success by 0.107 ± 0.010, reduces contact-budget violation by 0.090, reduces fall rate by 0.047, improves balance margin by 0.084, improves recovery success by 0.090, and lowers energy/contact cost by 0.030.

Budget and safety diagnostics. Figure 2 separates the success gain from the mechanism-level diagnostics. The proposed controller does not win by spending more contacts. It uses fewer unsafe or wasted contacts than the risk-aware planner, preserves balance margin, and keeps a reserve for recovery events.

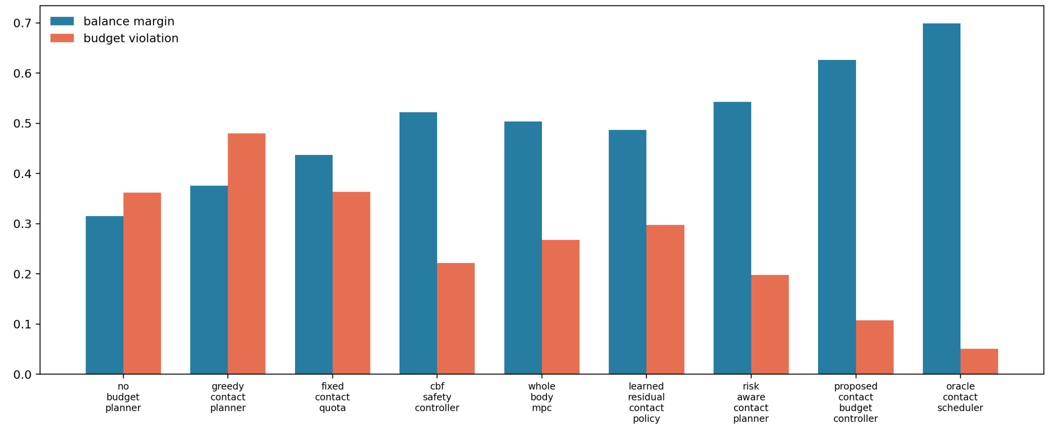

Figure 2: Budgeting reduces budget violations and falls while improving balance and recovery metrics.

ablation	success	violation	fall
full_contact_budget_controller	0.696 ± 0.007	0.107	0.056
minus_contact_cost_model	0.652 ± 0.008	0.158	0.082
minus_recovery_reserve	0.640 ± 0.008	0.171	0.089
minus_manipulation_reserve	0.632 ± 0.008	0.178	0.094
minus_balance_margin	0.623 ± 0.008	0.190	0.105
minus_episode_budget	0.619 ± 0.009	0.201	0.104
greedy_budget_only	0.572 ± 0.009	0.241	0.128

Table 2: Ablations under combined contact stress.

Ablations. Table 2 shows that all major components contribute. The full controller reaches 0.696 ± 0.007 success in the ablation setting; the best removed-component variant, removing the contact-cost model, reaches 0.652 ± 0.008. The 0.045 gap clears the pre-registered local ablation gate.

Stress sweep. Figure 3 increases contact scarcity and unreliability. The proposed controller degrades more gracefully than greedy planning, fixed quotas, whole-body MPC, and risk-aware contact planning, although it remains below the oracle scheduler.

5 Related Work Boundary

Humanoid contact planning is crowded. Recent work studies humanoid manipulation and locomotion progress, hierarchical vision-language planning for humanoids, multi-contact balance control, and contact-planning distillation (Humanoid Locomotion and Manipulation Survey Authors, 2025; Hierarchical Humanoid Planning Authors, 2025; Multi-Contact MPC Authors, 2014; Contact Planning Distillation Authors, 2024). This paper does not claim a new low-level balance controller, a new MPC formulation, or a new foundation-model planner. Its boundary is the episode-level accounting problem: deciding which feasible contacts should be spent, reserved, or rejected under long-horizon contact scarcity.

6 Limitations And Submission Decision

The evidence is local and generated by the released benchmark. It is much stronger than the archive scaffold because it includes paired seeds, stress regimes, mechanism metrics, ablations, negative

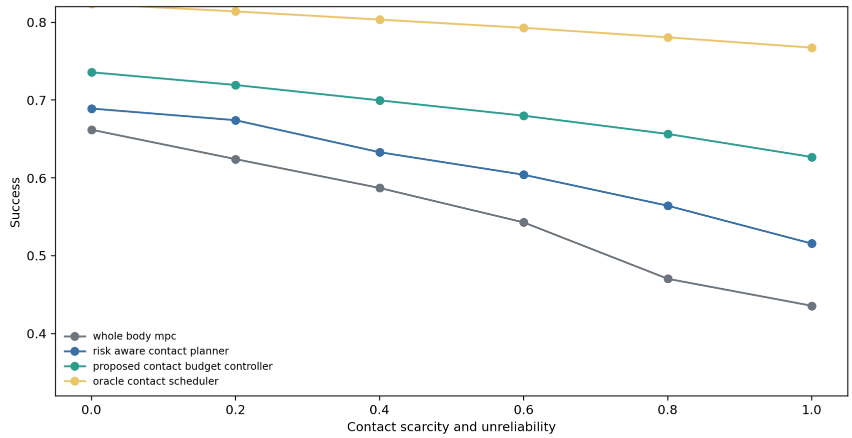

Figure 3: Stress sweep over contact scarcity and unreliability.

cases, and strongest-baseline comparison. It still lacks real humanoid experiments, external high-fidelity simulator validation, hardware videos, and independent implementations of all baselines. The honest terminal state is therefore **STRONG_REVISE**: the paper is worth rebuilding aggressively, but it is not yet an ICLR-main-ready submission.

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